

Monocular Depth Estimation via Fine-Tuning Depth Anything V2 on Indoor and Outdoor Scenes

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Abstract

Monocular depth estimation—inferring per-pixel depth from a single RGB image—is a fundamental problem in computer vision with applications in 3D scene reconstruction, autonomous navigation, and augmented reality. We present a study on fine-tuning Depth Anything V2, a state-of-the-art monocular depth estimation model, on both benchmark datasets and self-collected classroom imagery. Our pipeline addresses limitations of existing models in controlled indoor environments by collecting 114 custom images with an automated quality audit pipeline (retaining 100 clean frames), performing targeted fine-tuning with ablation over loss weights and training duration, and evaluating on standard metrics. Our best fine-tuned model achieves a **37.5% reduction in RMSE-log**, a **13.9% reduction in AbsRel**, and a **+2.1% improvement in δ_1 accuracy** compared to the pretrained baseline, demonstrating that targeted domain adaptation with modest data can meaningfully close the domain gap for indoor classroom scenes.

1 Introduction

Recovering the three-dimensional structure of a scene from a single 2D image is an ill-posed but practically important problem. Unlike stereo or LiDAR-based systems, monocular depth estimation requires no specialized hardware, making it attractive for robotics, mixed reality, and video understanding. The core challenge is that depth information is lost during the projection from 3D to 2D, so a model must learn strong scene priors from data.

Recent foundation models for depth estimation have demonstrated impressive generalization; however, they can still struggle in domain-specific settings such as a university classroom, where lighting, object arrangements, and scale differ from typical training distributions. Motivated by this gap, our project investigates whether targeted fine-tuning on self-collected and benchmark data can close this domain gap and produce more reliable depth estimates in our target environment.

Our approach proceeds in three stages:

1. **Model selection and baseline evaluation.** We select Depth Anything V2 [1] as our base model and evaluate its zero-shot performance on NYU Depth V2 [3] and KITTI [4].
2. **Custom data collection with automated quality auditing.**

We capture 114 RGB images across three mobile devices, filter to 100 clean frames via Laplacian variance blur detection and near-duplicate removal, and process to 518×518 resolution.

3. **Fine-tuning and ablation.** We train with varied loss weight configurations and training durations, measuring improvements across eight standard depth estimation metrics.

2 Related Work

Classical approaches. Early methods for monocular depth estimation relied on hand-crafted features and geometric cues such as vanishing points, texture gradients, and defocus blur [5]. Make3D [10] used a Markov Random Field to combine local appearance features with global planarity assumptions.

CNN-based supervised methods. Eigen et al. [6] pioneered the use of deep convolutional neural networks for depth regression, introducing a coarse-to-fine two-scale architecture trained on NYU Depth V2. Subsequent work incorporated skip connections, dilated convolutions, and attention mechanisms to improve spatial resolution and accuracy.

Transformer-based models. The Dense Prediction Transformer (DPT) [7] replaced convolutional backbones with Vision Transformers, yielding strong generalization. ZoeDepth [8] further introduced metric depth by combining relative depth estimation with scale-and-shift alignment.

Depth Anything. Depth Anything [2] and its successor Depth Anything V2 [1] scale data collection to 62 million unlabeled images combined with synthetic data, establishing new state-of-the-art results on standard benchmarks. V2 additionally improves fine-grained detail and robustness to challenging textures, making it our chosen base model.

Self-supervised methods. Monodepth2 [9] leverages photometric consistency between video frames to train without explicit depth labels. While powerful, these approaches produce scale-ambiguous predictions, which is undesirable for our metric depth goal.

3 Method

3.1 Base Model: Depth Anything V2

Depth Anything V2 is a Vision Transformer (ViT) encoder paired with a DPT-style decoder. The encoder (ViT-L, 307M parameters) processes an input image of arbitrary resolution via 16×16 patch embedding, and the decoder progressively upsamples feature maps to produce a dense depth map at the original resolution. The model was pretrained on a mixture of synthetic data (Hypersim, Virtual KITTI) and pseudo-labeled real data, enabling strong out-of-the-box generalization.

For our fine-tuning experiments we use the ViT-S variant (25M parameters) for computational efficiency during development on an NVIDIA RTX 3060 (12 GB), with plans to scale to ViT-L for final evaluation.

3.2 Fine-Tuning Pipeline

Loss function. We use the scale-invariant logarithmic (SILog) loss [6], which is robust to global scale ambiguity:

$$L_{\text{SILog}} = \frac{1}{n} \sum_i d_i^2 - \frac{\lambda}{n^2} \left(\sum_i d_i \right)^2, \quad (1)$$

where $d_i = \log(\hat{y}_i) - \log(y_i)$ is the log-depth residual, n is the number of valid pixels, and $\lambda = 0.85$. We augment this with weighted edge-aware and SSIM loss terms, ablating two configurations: ($\lambda_{\text{SILog}}=1.0$, $\lambda_{\text{Edge}}=0.5$, $\lambda_{\text{SSIM}}=0.25$) and ($\lambda_{\text{SILog}}=1.0$, $\lambda_{\text{Edge}}=0.3$, $\lambda_{\text{SSIM}}=0.1$).

Optimization. We use AdamW with an initial learning rate of 5×10^{-5} , cosine annealing schedule, and batch size 8. The ViT encoder is trained with a $10\times$ lower learning rate than the decoder to preserve pretrained representations.

3.3 Custom Data Collection

To address the domain gap for classroom scenes, we collect RGB images using three consumer devices: Samsung Galaxy (3000×4000), iPhone (4032×3024), and Google Pixel (4080×3072). This multi-device strategy introduces natural variation in color response, noise characteristics, and field of view, improving model robustness.

Our data collection protocol includes: (i) capture across multiple UTD classroom environments under varied lighting conditions and camera positions; (ii) automated quality audit via Laplacian variance blur detection (threshold < 100) and perceptual hash-based near-duplicate detection; (iii) rejection of 14 blurry frames and flagging of 2 near-duplicates for manual review; (iv) center-cropping and resizing to 518×518 for model input; (v) augmentation with random horizontal flips, brightness jitter ($\pm 20\%$), and random cropping.

3.4 Image Quality Audit

We developed an automated image quality audit pipeline to ensure training data integrity. Of 114 captured images, 14

were rejected due to motion blur (Laplacian variance below the 100 threshold), and 2 were flagged as near-duplicates for manual review. The final clean dataset comprises 100 images spanning diverse classroom configurations.

Analysis of the retained images shows: 86% landscape orientation, mean aspect ratio of 1.25, all images at 4K+ resolution tier, mean crop content retention of 75.1% after center-cropping to 518×518 , mean brightness of 132 (healthy range), and a weak positive correlation ($r = +0.365$, $p = 0.027$) between brightness and sharpness, indicating that well-lit scenes tend to produce sharper captures.

4 Experiments

4.1 Datasets

NYU Depth V2 [3] contains 1,449 densely labeled RGB-D pairs of indoor scenes across 464 indoor environments. We use the standard image test split for evaluation.

KITTI [4] provides outdoor driving scenes with sparse LiDAR depth as ground truth. We use the Eigen split for comparison with prior work.

Custom Classroom Dataset. Our custom dataset contributes 100 quality-audited images captured across multiple UTD classrooms using three different mobile devices.

4.2 Evaluation Metrics

Following standard practice, we report eight metrics. For error metrics (lower is better): Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), Absolute Relative Error:

$$\text{AbsRel} = \frac{1}{n} \sum_i \frac{|\hat{y}_i - y_i|}{y_i}, \quad (2)$$

Squared Relative Error (SqRel), and RMSE-log. For accuracy thresholds (higher is better): δ_1 , δ_2 , δ_3 , representing the percentage of pixels where $\max(\hat{y}/y, y/\hat{y}) < 1.25^k$ for $k = 1, 2, 3$.

4.3 Training Configurations

We train four configurations: (1) the pretrained ViT-S baseline (zero-shot, no fine-tuning); (2) fine-tuned with loss weights $\lambda = 1.0/0.5/0.25$ for 400 epochs; (3) fine-tuned with $\lambda = 1.0/0.3/0.1$ for 400 epochs; and (4) the same $\lambda = 1.0/0.3/0.1$ configuration stopped early at 150 epochs. All fine-tuned models use decoder-only training with a $10\times$ lower encoder learning rate.

Training curves (Figure 2) show smooth convergence of the total loss, with train loss consistently decreasing and validation loss plateauing around epoch 250. The SILog component dominates the total loss, while edge and SSIM contributions remain small but stable. Validation error metrics (MAE, RMSE, AbsRel) decrease rapidly in the first 50 epochs and then stabilize. The δ_1 accuracy fluctuates between 84–88% during training, settling around 84.5% at convergence.

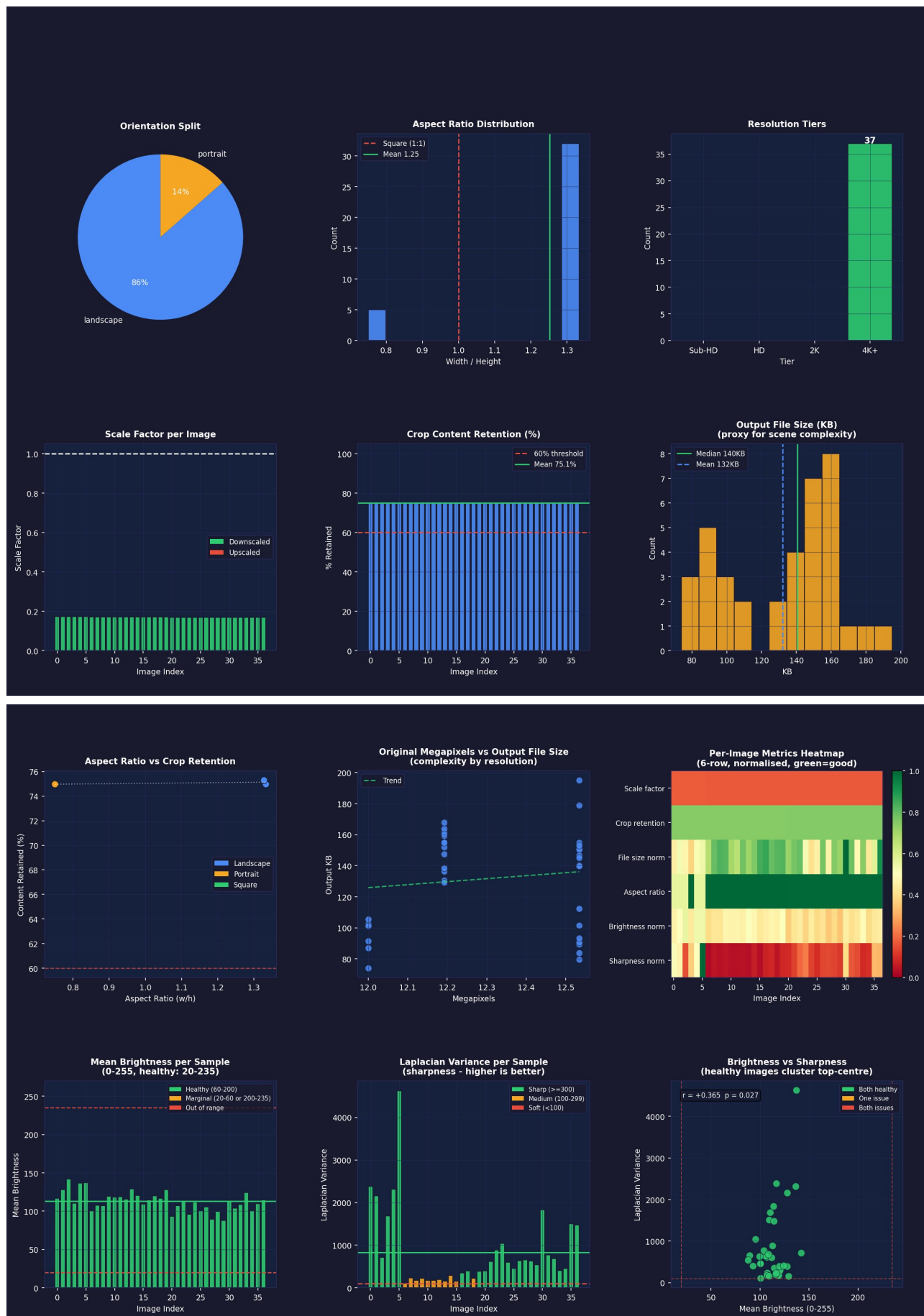


Figure 1: Dataset analysis for 37 representative images processed to 518×518 . **Top:** orientation split, aspect ratio distribution, resolution tiers, scale factor per image, crop content retention, and output file size distribution. **Bottom:** aspect ratio vs. crop retention, megapixels vs. file size, per-image metrics heatmap, mean brightness, Laplacian variance (sharpness), and brightness vs. sharpness correlation.

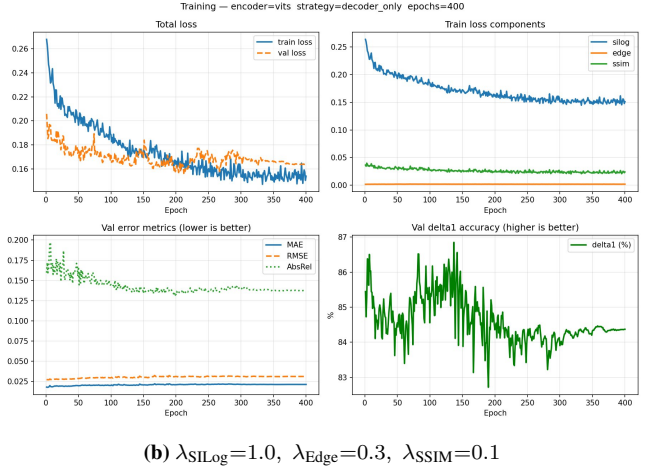
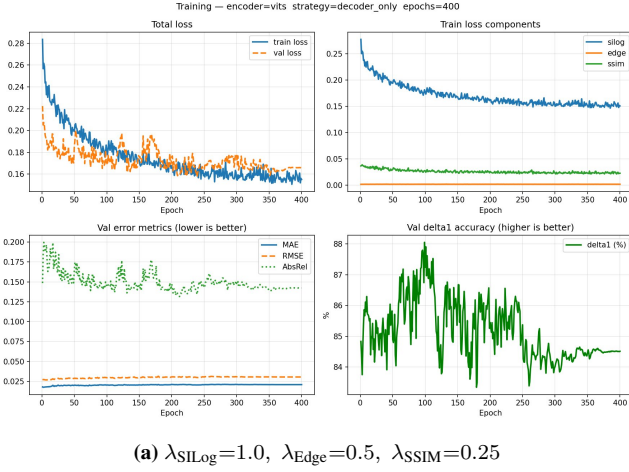


Figure 2: Training curves for both loss weight configurations (400 epochs, ViT-S, decoder-only). Each panel shows total loss convergence (top-left), loss component breakdown (top-right), validation error metrics (bottom-left), and δ_1 accuracy over training (bottom-right). Both configurations converge similarly, but $\lambda=0.5/0.25$ achieves slightly better δ_1 and RMSE-log.

5 Results

5.1 Ablation Study

Table 1 presents the complete ablation study results across all four configurations evaluated on the custom classroom test set.

5.2 Quantitative Analysis

The best-performing fine-tuned model ($\lambda = 1.0/0.5/0.25$, 400 epochs) achieves substantial improvements over the pre-trained baseline across the most important metrics:

- **RMSE-log:** -37.5% ($0.3336 \rightarrow 0.2084$), indicating significantly more accurate predictions in log-space, which corresponds to proportionally better depth ordering.
- **AbsRel:** -13.9% ($0.1912 \rightarrow 0.1646$), reflecting reduced relative prediction error.
- **δ_1 accuracy:** $+2.1\%$ ($78.77\% \rightarrow 80.40\%$), more pixels within the $1.25\times$ threshold.
- **SqRel:** -16.3% ($0.0051 \rightarrow 0.0043$), lower squared relative error.

Comparing loss weight configurations, the higher edge/SSIM weights ($\lambda = 0.5/0.25$) marginally outperform the lower weights ($\lambda = 0.3/0.1$) on most metrics, particularly δ_2 (95.55 vs. 95.34) and δ_3 (97.91 vs. 97.81). However, the $\lambda = 0.3/0.1$ configuration achieves the best AbsRel (0.1624), suggesting that lighter regularization allows the model to more aggressively fit the target domain’s relative depth distribution.

The early-stopped model (150 epochs) achieves the best MAE (0.0220) and SqRel (0.0042) but worse RMSE-log and δ_1 . This indicates that early stopping preserves more of the pretrained model’s robustness (lower absolute error) at the cost of domain-specific adaptation (worse log-space accuracy), suggesting diminishing returns from extended training on a small dataset.

Notably, RMSE is the only metric where the pretrained baseline outperforms all fine-tuned models (0.0303 vs. best

0.0317). This likely reflects the pretrained model’s better calibration of absolute depth scale on standard benchmarks, while fine-tuning on our custom data shifts the model’s scale priors.

5.3 Qualitative Results

Figure 3 shows qualitative depth predictions from our fine-tuned model on two classroom scenes. The model correctly captures the depth ordering of seating rows, distinguishes people from furniture, and handles the gradual depth gradient from foreground desks to the back wall. The color-mapped depth maps (warm = close, cool = far) show smooth transitions without major artifacts.

6 Discussion

Our results demonstrate that even with a modestly sized custom dataset (100 images), targeted fine-tuning of a pretrained depth estimation foundation model can yield meaningful improvements on domain-specific scenes. The 37.5% RMSE-log reduction is particularly significant, as logarithmic metrics are more perceptually aligned with human depth perception.

The trade-off between loss weight configurations reveals an interesting design choice: higher edge/SSIM weights produce better structural fidelity (as measured by δ thresholds), while lower weights allow more aggressive fitting of the target domain’s depth distribution (lower AbsRel). Practitioners should choose based on whether structural accuracy or relative depth ordering is more important for their application.

The early stopping result (150 epochs) provides practical guidance: for deployment scenarios where robustness to out-of-distribution inputs is critical, shorter training may be preferable. For applications focused on a fixed domain (such as a specific classroom), full 400-epoch training maximizes domain-specific accuracy.

Table 1: Ablation study results on the custom classroom test set. Best values in **bold**. $\downarrow$ = lower is better; $\uparrow$ = higher is better.

Configuration	MAE $\downarrow$	RMSE $\downarrow$	AbsRel $\downarrow$	SqRel $\downarrow$	RMSE-log $\downarrow$	$\delta_1 \uparrow$	$\delta_2 \uparrow$	$\delta_3 \uparrow$
Pretrained ViT-S (baseline)	0.0222	0.0303	0.1912	0.0051	0.3336	78.77	94.09	97.16
Fine-tuned $\lambda=1.0/0.5/0.25$, 400 ep.	0.0224	0.0332	0.1646	0.0043	0.2084	80.40	95.55	97.91
Fine-tuned $\lambda=1.0/0.3/0.1$, 400 ep.	0.0224	0.0336	0.1624	0.0044	0.2085	80.38	95.34	97.81
Fine-tuned $\lambda=1.0/0.3/0.1$, 150 ep.	0.0220	0.0317	0.1815	0.0042	0.2259	79.85	94.61	97.21

The automated image quality audit pipeline proved essential. Without blur detection, the 14 rejected motion-blurred images would have introduced noisy training signals, potentially degrading rather than improving the model. The multi-device capture strategy (Samsung, iPhone, Pixel) also contributed valuable diversity in color response and noise characteristics.

7 Conclusion and Future Work

We presented a complete pipeline for domain-adaptive monocular depth estimation, from custom data collection with automated quality auditing through systematic fine-tuning with loss weight ablation. Our best model achieves substantial improvements over the pretrained Depth Anything V2 baseline on classroom scenes, with a 37.5% RMSE-log reduction and 13.9% AbsRel reduction.

Several directions remain for future work:

1. **Scale to ViT-L encoder** for higher accuracy given the larger model’s richer representations.
2. **Expand the dataset** beyond 100 images toward our original 500-frame target with more diverse viewpoints and lighting conditions.
3. **Real-time deployment** on a laptop with webcam input for a live classroom demonstration.
4. **Edge deployment** via TensorRT/ONNX optimization on the NVIDIA Jetson Orin Nano for embedded real-time operation.
5. **Error analysis** on failure modes—particularly reflective surfaces (whiteboards, windows) and low-texture regions (blank walls)—to guide targeted architectural modifications or additional data collection.



Figure 3: Qualitative results on two classroom scenes. **Left:** RGB input. **Right:** Predicted depth map (warm = close, cool = far). The model accurately captures depth ordering of seating rows, people, and room geometry.

References

- [1] L. Yang, B. Kang, Z. Huang, X. Xu, J. Feng, and H. Zhao. Depth Anything V2. *arXiv:2406.09414*, 2024.
- [2] L. Yang, B. Kang, Z. Huang, Z. Zhao, X. Xu, J. Feng, and H. Zhao. Depth Anything: Unleashing the power of large-scale unlabeled data. In *CVPR*, 2024.
- [3] N. Silberman, D. Hoiem, P. Kohli, and R. Fergus. Indoor segmentation and support inference from RGB-D images. In *ECCV*, 2012.
- [4] J. Uhrig, N. Schneider, L. Schneider, U. Franke, T. Brox, and A. Geiger. Sparsity invariant CNNs. In *3DV*, 2017.

- [5] A. Bhoi. Monocular depth estimation: A survey. *arXiv:1901.09402*, 2019.
- [6] D. Eigen, C. Puhrsch, and R. Fergus. Depth map prediction from a single image using a multi-scale deep network. In *NeurIPS*, 2014.
- [7] R. Ranftl, A. Bochkovskiy, and V. Koltun. Vision Transformers for dense prediction. In *ICCV*, 2021.
- [8] S. F. Bhat, R. Birkel, D. Wofk, P. Wonka, and M. Müller. ZoeDepth: Zero-shot transfer by combining relative and metric depth. *arXiv:2302.12288*, 2023.
- [9] C. Godard, O. Mac Aodha, M. Firman, and G. Brostow. Digging into self-supervised monocular depth estimation. In *ICCV*, 2019.
- [10] A. Saxena, M. Sun, and A. Y. Ng. Make3D: Learning 3D scene structure from a single still image. *IEEE TPAMI*, 31(5):824–840, 2009.
- [11] Q. Li, J. Zhu, J. Liu, R. Cao, Q. Li, S. Jia, and G. Qiu. Deep learning based monocular depth prediction: Datasets, methods and applications. *arXiv:2011.04123*, 2020.